Abdelrahman Khalifa

Software Engineer | Backend, Python, Algorithms

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SUMMARY

Software engineer with a strong algorithms background: Codeforces Expert, 4-time ECPC finalist and mentor in the local ICPC community. Built a Node.js/MongoDB REST backend and a Python app at EKSON, and got 11 PRs merged upstream as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting). Comfortable with SQL and data modeling, writing tests, and learning new stacks quickly.

EXPERIENCE

Embedded Software Engineer | **EKSON** | **Aug 2025 – Present**

Software for 3-DoF and 6-DoF pneumatic motion simulators: backend, desktop app, firmware and CI

- Built the backend for remote operation and firmware updates: a Node.js/Express REST API with MongoDB that the desktop app polls to relay commands, store session and telemetry data, and deliver new firmware; GitHub Actions CI.
- Developed the Python (PyQt5) control app; used asyncio to wait for device and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS, with PID control, safety stops and a UART protocol.

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

Open-source project: Printers page for COSMIC Settings, in Rust; passed the final evaluation | github.com/Abd002/GSOC-2026

- Designed and built the **Printers page for COSMIC Settings** as 5 Rust crates, with the backend as a separate service behind a varlink API so a crash cannot take down Settings (cosmic-settings #2182, in review).
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with CI across architectures, added API bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

Contributor (all merged) — Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | **Mar 2025 – Dec 2025**

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 (#86977); Boa: faster `JsStr::code_points()` for Latin1 strings (PR #4553); Rust Clippy: fixed `ref_as_ptr` false positives (PR #16214).

PROJECTS

- **Inventory Management System** (C#, SQL Server): stock, orders and deliveries with role-based access; MVVM and the Observer pattern; database audit log built with SQL Server triggers that record the logged-in user through the session context. GitHub
- **Sports Management Web App** (team; ASP.NET Core MVC, MS SQL, Entity Framework): designed the relational schema and wrote the backend and account-management logic for role-based dashboards. GitHub
- **Database Systems, CMU 15-445** (C++17, SQL): thread-safe Count-Min Sketch for the BusTub database (per-row locks, merge, top-k); SQL analytics in DuckDB. GitHub
- **Function Solver and Plotter** (Python): solves and plots functions with SymPy and Matplotlib; input validation and end-to-end tests with pytest-qt. GitHub

TECHNICAL SKILLS

Languages: Python, JavaScript (Node.js), SQL, C++, C, Rust, C#, Java

Backend & data: REST API design, Express, MongoDB, SQL Server, DuckDB, data modeling, ASP.NET Core MVC, Entity Framework

Computer science: Data structures and algorithms, complexity and optimization, concurrency, operating systems, database internals

Tools: Git/GitHub, GitHub Actions CI, pytest, Linux, debugging, code review

ACHIEVEMENTS

- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024 (Egyptian Collegiate Programming Contest).
- Mentor in the local ICPC community: taught problem solving in C++ to 20+ colleagues. Treasurer, IEEE Student Branch (budget of more than $10k).

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors (84.6%).